1. Project Title: Connect 4 AI Player

2. Team:

- Nathan Zhang:
  Lead for Hardware - Get the LED matrix to connect seamlessly with the human player input. Collect data for the Computer Vision model.
- Kristian Prifiti:
  Lead for Software - Train the Computer Vision model. CAD the base for the project.

3. Motivation:
We wanted to create a fun, interactive game you can play solo without needing to stare at a screen. Connect 4 is simple, fast, and familiar, making it perfect for a physical, hands-on version powered by embedded AI. By combining an LED matrix, a camera, and onboard decision-making, we're building an engaging tabletop experience that blends classic gameplay with modern hardware.

4. Goal:
The goal is to build a fully functional Connect 4 system where a human player competes against an AI opponent. The 6×7 game grid will be displayed on an LED matrix. A camera mounted above the board will continually read the LED matrix to detect the current board state. The user will input their moves through buttons. Once the human move is registered, the AI will compute and display its move on the LED matrix. The final system will run autonomously and play full games of Connect 4.

5. Methodology:

- Build the base game
  - Build a working version of the LED-matrix with buttons where we can display a current state of the game.
- Collect data
  - Take pictures of different states with the esp 32 camera module.
  - Label the data as a 6x7 matrix where 0 represents empty cells, 1 represents cells occupied by player 1 and 2 represents the cells occupied by player 2.
- Train Vision Model
  - Train a model that can take a picture of the LED matrix and convert it into a 6x7 matrix with 0s, 1s, and 2s to represent the current state of the game.
- AI player
  - Firstly we will use a pre-build connect 4 AI model (or something similar) to decide on a next move for the AI player.
  - Later on we want to train our own model using reinforcement learning.
- Finishing touches
  - 3D print a custom base for the LED matrix and the camera mount.
  - Make sure the gameplay is smooth

6. Project Components:
The project will be built using several key hardware and software components. On the hardware side, the central controller will be a Seeed Studio XIAO, which includes an integrated camera module. The game board will be on a 32x32 RGB LED matrix (the actual game is 6x7), serving as the primary visual display. User input for the human player will be handled by a simple set of push buttons connected to the XIAO's GPIO pins. For software, the XIAO will be programmed using C++. The computer vision model will be developed and trained in Google Colab.

7. Testing and Evaluation:
For the vision system, we will measure its accuracy in correctly identifying the state of every cell on the board; a good outcome will be an accuracy of over 90% on a held-out test set of board images. System performance will be measured by its latency, or the time from when a human move is made to when the AI's counter-move is displayed on the board. A good outcome would be a latency of under 2 seconds to ensure a fluid and responsive user experience.

8. Deliverables:
Baseline1: A functional hardware setup where the XIAO microcontroller can successfully control the RGB LED matrix to display custom patterns and colors, confirming that the hardware interface is working correctly.

Baseline2: A complete two-player Connect 4 game running on the system. Two human players can take turns making moves using a physical button interface, with the LED matrix accurately reflecting the game state and correctly identifying and announcing win, loss, or draw conditions.

Baseline3: A functional single-player mode that incorporates the vision system. The integrated camera will be fully operational, and the machine learning model will be able to accurately recognize the state of the game board from the camera feed. Based on this visual input, the AI opponent will be able to analyze the board and make a valid move in a reasonable amount of time (<2 seconds).

Reach Goals - Extra Credit: (justify why this should not be in the baseline)
Advanced Reinforcement Learning Agent: Train a sophisticated AI using deep reinforcement learning techniques that can learn strategies to play competitively. Justification: This requires significant computational resources and time for training, which may be beyond the scope and timeline of the project.

9. Overall Timeline: (in 3 days sprints)

Day 1: Finishing the hardware component, make sure the LED matrix works and can get input from the buttons. Collect and label data that will be used for the Computer Vision component.
Day 2: Train the computer vision model, make sure it is accurate at detecting game state. CAD a base for the game so there is enough time for it to get printed.
Day 3: Finish the interactive gameplay. Train the Reinforcement learning model. Place everything to the 3D printed base.